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26 July 1988
SUPERSEDING
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MILITARY SPECIFICATION

LINER, FIELD PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers three sizes of waterproof field pack liners (see 6.1).

1.2 Classification. The liners shall be of one type and three sizes (see 6.2).

Schedule of Sizes

Size 1 - 17-3/4 inches high by 21-1/2 inches in circumference.
Size 2 - 27 inches high by 39 inches in circumference.
Size 3 - 30 inches high by 50-1/2 inches in circumference.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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SPECIFICATIONS

FEDERAL

A-A-203	-	Paper, Kraft, Untreated
O-I-503	-	Insect Repellent, Clothing and Personal Application
V-T-285	-	Thread, Polyester
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-5040	-	Cord, Nylon
MIL-C-21108	-	Cloth, Nylon, Raft Bottom
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43473	-	Cloth, Coated, Nylon, Polyurethane Coated

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus
D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Guide sample. Guide samples when furnished are solely for guidance and information (see 6.3). Variations from the document may appear in the sample, in which case the document shall govern.

* 3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.5).

3.3 Materials.

* 3.3.1 Polyurethane coated cloth. The polyurethane coated cloth shall conform to the following requirements, or as an alternate, shall conform to type I of MIL-C-43473.

3.3.1.1 Base cloth. The base cloth shall conform to type I of MIL-C-21108. The color shall be Olive Green 106.

3.3.1.2 Coating compound. The coating compound shall be a composition of polyurethane suitably compounded and pigmented to meet the requirements herein. If the polyurethane is plasticized, only phthalate or phosphate ester plasticizers shall be used. The color shall match Olive Green 207.

* 3.3.1.3 Coated cloth. The cloth shall be coated on one side only with the coating specified in 3.3.1.2 and shall conform to all the requirements of table I when tested as specified in 4.4.1.

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ll be made to 4.4.2 In-process inspection. Inspection of sub-assemblies sha

TABLE I. Physical requirements of coated cloth

Characteristic	Requirement	
	Minimum	Maximum
Weight, ounces/square yard	4.0	4.5
Breaking strength, pounds:		
Warp	115	--
Filling	115	--
Tearing strength, grams:		
Warp and filling	1200	--
Hydrostatic resistance, pounds/square inch:		
Initial	150	--
After accelerated weathering	50 1/	--
After abrasion	50	--
After adhesion of coating	50	--
After high humidity	50	--
Colorfastness after high humidity	No appreciable change 2/	
Stiffness, centimeters (warp only)	--	10.0
Adhesion of coating, pounds/2 inch width	7.0	--
Blocking, scale rating	--	No. 2
Resistance to insect repellent	pass 3/	--

1/ The coating shall not become stiff and brittle nor soft and tacky and there shall be no evidence of cracking or crazing under visual examination after 100 hours exposure.

2/ "Appreciable change in color" means a change which is immediately noticeable in comparing the test specimen with the original sample for comparison. If closer inspection or a change of angle of light is required to make apparent a slight change of color, the change is not considered appreciable.

3/ The coated cloth shall show no lifting of the coating, no tackiness, no pick off of the coating, and no adherence of the coating to itself greater than scale rating No. 3 (slight blocking).

3.3.2 Adhesive. Adhesive for laying of turned seams, bonding of strapping over seams, sealing the bartack stitching, and cementing the hem shall be a urethane base solvent cement activated by addition of an isocyanate type accelerator. The cured adhesive shall meet the blocking resistance requirement of 3.8.2, and the strapping adhesion requirement of 3.8.3.

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3.3.3 Cord, nylon. The cord shall conform to type IIA of MIL-C-5040. The color shall be Olive Drab 7.

* 3.3.4 Thread. The thread used for stitching shall conform to type I or II, class 1, subclass B, size B, Olive Drab S-1 C.A. 66022 of V-T-285. The thread shall show good fastness to light.

* 3.3.5 Strapping. The strapping used to strap all seams shall be made of the coated cloth specified in 3.3.1.3. The strapping shall be bias cut and after being adhered to the liner shall measure not less than 1 inch wide. The strapping suppliers splice or fabricators overlap splice may be used, limited to one such splice per seam. The bottom seam shall be completed by overlapping the ends 1 ± 1/4 inch. No other splicing or overlapping is authorized.

3.4 Dusting powder. When used, the dusting powder shall be whiting, talc or other finely divided mineral material which does not support mildew growth.

3.5 Design. The design of the liners shall be in accordance with figure 1. The coated side of the cloth shall be on the outside of the liner.

3.6 Construction.

3.6.1 Stitches, seams, and stitching. All stitches, seams and stitching shall conform to FED-STD-751. Stitch type 301 shall be used for the hem and stitch types 301 and 401 shall be used for all other seams, using not less than 8 or more than 10 stitches per inch. Whenever stitch type 401 is used, the looper (i.e. the under thread) shall be on the inside of the liner. The needle shall be the smallest size which can be used with the specified thread in order to minimize the size of the hole made in the fabric by the needle. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operation is prohibited.

3.6.1.1 Type 301 stitching. Ends of all stitchings shall be backstitched or overstitched not less than 1 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewed.

3.6.1.1.1 Repairs of type 301 stitching.

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching.

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b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue over the defective area, and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner.

When making the above repairs in a. and b. above, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 401 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.6.1.1.1 a. and 3.6.1.1.1 b. except that the stitching break shall be repaired by stitching 1-1/2 inches in back of, and in front of the break.

3.6.2 Bartack. Bartack shall be 1/2 ± 1/16 inch long and 1/8 ± 1/32 inch wide and shall contain 28 stitches. The bartack shall be free from thread breaks and loose or tight stitching.

3.6.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.7 Manufacturing operations requirements. The use of drill holes is prohibited in the construction of the liners. The tie cord attachment shall be reinforced with a 2 ± 1/8 inch diameter patch of coated cloth material which shall be cemented to the inside of the liner with the adhesive specified in 3.3.2. The bartack shall be through all layers of material. Liners of each size shall be made with the coating on the outside of the liner.

3.7.1 Liner construction. The liner shall be constructed from the coated cloth specified in 3.3.1.3. All joining seams shall be shingled and single stitched (seam type SSa-1), laid in cement, turned to one side and cemented down forming a 3/8 inch seam on the inside of the liner. Strapping specified in 3.3.5 shall then be cemented in a centered position over the stitched side seam and bottom seam with the coated side of strapping contacting the inside of the seam. Cementing shall be thorough with no loose edges and the cement shall not extend more than 3/8 of an inch on either side of the strapped seam. The strapped seams shall meet the hydrostatic requirement of 3.8.1.

3.7.2 Hem. The hem shall be formed by turning the top of the liner 1/2 ± 1/8 inch. Stitch 1/8 ± 1/16 inch from the raw edge using stitch type 301 and seam type EFa-1. The finished hem shall measure 1/2 ± 1/8 inch. Application of sealant or strapping to the hem is not required.

3.7.2.1 Hem (alternate construction). Alternatively the hem may be formed by turning the top of the liner 1/2 inch and firmly cementing using adhesive specified in 3.3.2. The finished hem shall measure 1/2 ± 1/8 inch.

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3.7.3 Tie cord. The tie cord shall be made of the cord specified in 3.3.3. The cord shall be bartacked to the outside of the liner, through the strapped side seam and through the reinforcement patch which is cemented to the inside of the liner. The cord shall be 1-7/8 ± 1/8 inch from the top edge of the liner. The cord shall be bartacked with a 1/2 inch bartack as specified in 3.6.2 (see figure 1). The bartack shall be positioned parallel to the cord and centered in the cord which is perpendicular to the back seam. The bartack shall be made waterproof on the inside with the adhesive specified in 3.3.2. When attached to the liner, the tie cord for size No. 1 liner shall measure not less than 20 inches, the tie cord for size No. 2 liner shall measure not less than 38 inches, and the tie cord for size No. 3 liner shall measure not less than 60 inches when measured in the relaxed state; and the ends of the tie cord shall not be uneven in length by more than 3 inches. The ends of the cord shall be seared or fused to prevent unraveling.

3.8 Seam performance.

3.8.1 Hydrostatic resistance. The joining seams and the area where the tie cord is bartacked to the liner shall withstand a hydrostatic pressure of 25 centimeters for 1 minute, without leaking when tested as specified in 4.4.6.

3.8.2 Blocking. Resistance of seams to blocking shall not exceed No. 3 when tested as specified in 4.4.6.

3.8.3 Adhesion. There shall be no more than 1/8 inch peeling when strapping adhered to the liner is tested for adhesion as specified in 4.4.6.

3.9 Measurements.

3.9.1 Size No. 1. The finished liner shall be 17-3/4 ± 1/2 inches in height and 21-1/2 ± 1/2 inches in circumference.

3.9.2 Size No. 2. The finished liner shall be 27 ± 1/2 inches in height and 39 ± 1/2 inches in circumference.

3.9.3 Size No. 3. The finished liner shall be 30 ± 1/2 inches in height and 50-1/2 ± 1 inch in circumference.

3.10 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written document and the figure, the document shall govern.

3.11 Label. Each liner shall have identification and instruction labels located on the outside of the liner adjacent to the tie cord (see figure 1). The labels shall conform to type III or IV, class 5 of DDD-L-20. The instruction label shall be printed below the identification label in letters 1/8 inch high. The contents of the instruction label shall be as shown in figure 2. The overall size of the instruction label shall be a minimum of 6 inches wide by 7 inches long.

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when exposed to high power densities, thus the effective optical density at a

3.12 Workmanship. The liners shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3, 4.4.4, and 4.4.5 and tested as specified in 4.4.1 and 4.4.6, as applicable. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

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* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition to the quality assurance provisions of the subsidiary specifications and drawings, testing shall be performed on the coated cloth for the characteristics specified in table II. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table II shall be followed. The requirements for each characteristic listed in table II are applicable to the sample unit, except "Weight" which is applicable to the lot average. The lot size for sampling purposes shall be expressed in units of 1 yard (full width) of coated cloth. The sample unit shall be 2 yards of coated cloth. The inspection level shall be S-2. For all characteristics except "Weight", the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. For the "Weight" characteristic, there shall be no evidence of failure of the average of all sample units to meet the requirement specified.

TABLE II. Component tests

Characteristic	Requirement paragraph	Test method
Weight	3.3.1.3	5041
Breaking strength	3.3.1.3	5100
Tearing strength	3.3.1.3	ASTM D 1424 1/
Hydrostatic resistance:		
- Initial	3.3.1.3	5512 2/
- After accelerated weathering	3.3.1.3	5804 3/ & 5512 2/
- After abrasion	3.3.1.3	5302 4/ & 5512 2/
- After adhesion of coating	3.3.1.3	5972 & 5512 2/
- After high humidity	3.3.1.3	4.5.1
Colorfastness after high humidity	3.3.1.3	4.5.1
Stiffness (warp only) at standard conditions	3.3.1.3	5204 5/
Adhesion of coating	3.3.1.3	5970
Blocking	3.3.1.3	5872 6/

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TABLE II. Component tests (cont'd)

Characteristic	Requirement paragraph	Test method
Resistance to insect repellent	3.3.1.3	4.5.2

1/ Except results shall be reported to the nearest 10 grams.

2/ The water pressure shall be applied only to the uncoated side of the coated cloth.

3/ The specimens, with the coated side facing the light source, shall be exposed in the weatherometer for 100 hours. At the end of the exposure period, the specimens shall be visually examined to determine if the coating has become stiff and brittle or soft and tacky and then tested for hydrostatic resistance.

4/ A solid rubber diaphragm 0.030 ± 0.010 inch thick (no metallic contact) shall be used. The abradant shall be the coated side of the coated cloth. The coated side of the specimen shall be abraded 1000 double strokes and then tested for hydrostatic resistance with the abraded portion of the test specimen centered in the hydrostatic test area.

5/ Except results shall be reported to the nearest 0.1 centimeter.

6/ One specimen from each sample unit shall be tested.

* 4.4.1.1 Certification. Components and materials listed below may be accepted on the basis of a supplier's certificate of compliance.

Component	Requirement reference
Coating compound (each batch)	3.3.1.2
Adhesive and sealing compound	3.3.2
Dusting material	3.4

* 4.4.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

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* 4.4.3 End item visual examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of liners of one size only. The sample unit shall be one liner. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Design	Not as specified.	X	
Construction	Coated side of fabric on inside of liner.	X	
Liner body and bottom	Any part of liner spliced, i.e., made with more than one piece of fabric.	X	
Seams	Any part of seams or strapping not securely cemented.	X	
	Any tunnel or air pocket on strapping.	X	
	Strapping less than 1 inch wide but not less than 3/4 inch wide.		X
	Strapping less than 3/4 inch wide.	X	
	Strapping misplaced, exposing stitching.	X	
	Any individual seam not strapped with one continuous piece of strapping excepting the strapping suppliers splice.		X
	Cement extending more than 3/8 inch on either side of strapped seams.		X
	Bottom seam strapping ends overlapped less than 3/8 inch long.		X
Hem	Any unsealed area on cemented hems more than 1/4 inch in depth or length.		X
	Twisted or irregular in width causing visible unevenness of edges.		X
Tie cord	Cut, torn, frayed, or spliced.		
	Ends uneven in length by more than 3 inches.		X
	Set to liner less than 1-3/4 inches or more than 2 inches from top edge.		X
	Any end of tie cord not seared or fused.		X

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Examine	Defect	Classification Major	Minor
Tie cord reinforcing patch	Patch not securely cemented.	X	
	Patch on the outside of liner.	X	
Bartack	Number of stitches or length of bartack less than specified.		X
	Not waterproofed on inside.	X	
	Stitching loose or broken.	X	
	Tie cord bartack not through side seam and reinforcement piece.	X	
	Tie cord bartack not positioned horizontally on seam.		X
Liner for material defects in coated fabric	Any hole, cut, or tear.	X	
	Fabric or coating defects:		
	- seriously affecting serviceability.	X	
	- affecting serviceability or appearance but not seriously.		X
Liner for workmanship defects	Any needle chews, mends, or patches (other than reinforcement patch).	X	
	Peeled or blistered areas on any seam.	X	
	Any drill hole in fabric.	X	
	Grease or oil stains clearly noticeable.		X
	Thread ends not trimmed throughout.		X
Components and assembly	Any required operation omitted or not as specified:		
	- seriously affecting serviceability or appearance.	X	
	- affecting serviceability or appearance but not seriously.		X
	Any required component or part of liner omitted.	X	
	Seams or strapping poorly cemented or cement not evenly applied, resulting in peeled, blistered, or unsealed areas.	X	
	Chain portion of stitching (on stitch type 401) exposed on outside of liner.	X	
	Dusting powder omitted from seam areas or improperly applied, causing adhesion of fabric to seam area or resulting in tacky sealant.	X	

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Examine	Defect	Classification Major	Minor
Identification and instruction labels	Omitted, incorrect, illegible, misplaced, wrong size, not in proper location, or not accomplished in the specified manner.		X

4.4.4 End item thru-lighting examination. The liners shall be examined by means of a thru-lighting device similar to that shown in figure 3 as required. The thru-lighting shall be performed in a darkened area. A darkened area is defined as any area where the horizontal surface of the liner to be inspected (liner in place over test fixture without test light on) shall be externally illuminated with 20 ± 5 foot candles of natural or artificial or a combination of natural and artificial light (see 6.4). The liner shall be thru-light examined with the coated side of the liner facing out. The viewer's eye level shall be kept perpendicular to the area of the liner being examined. The presence of any hole or pinhole shall be classified as a defect. The lot size shall be expressed in units of liners. The sample unit shall be one liner. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.4.5 End item dimensional examination. The end items shall be examined for the defects listed below. The lot size shall be expressed in units of liners of one size only. The sample unit shall be one liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect Size 1	Size 2	Size 3
Height of liner	Less than 17-1/4 inches. More than 18-1/4 inches.	Less than 26-1/2 inches. More than 27-1/2 inches.	Less than 29-1/2 inches. More than 30-1/2 inches.
Circumference of liner	Less than 21 inches. More than 22 inches.	Less than 38-1/2 inches. More than 39-1/2 inches.	Less than 49-1/2 inches. More than 51-1/2 inches.
Top of edge of liner	Hem less than 3/8 inch or more than 5/8 inch in width.		
Tie, cord	Not specified length. 1/		

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1/ The liner shall be laid out smooth and flat on a table with the tie cord fully extended. A weighted clamp approximately 3 to 4 ounces shall be securely attached to one end of the tie cord. The tie cord shall be fully extended by pulling on the other end of the tie cord with enough force to move the weighted clamp very slowly along the surface of the table. The tie cord shall be straight and parallel to the surface of the table while measuring.

* 4.4.6 End item testing. The end items shall be tested for the characteristics specified in table III. The lot size shall be expressed in units of liners of one size only. The sample unit shall be two liners. The inspection level shall be S-1 and for each characteristic the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Hydrostatic resistance	3.8.1	4.5.3
Seam Blocking	3.8.2	4.5.4
Adhesion of cemented strapping	3.8.3	4.5.5

4.4.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior)	Omitted; incorrect; illegible; of improper size location, sequence, or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Improperly folded liners, inadequately tied bundles, contents improperly covered with paper.

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Examine	Defect
Contents	Number of bundles is more or less than required. Number of liners per bundle is more or less than required. 1/

1/ For this defect, one bundle from each shipping container in the sample shall be examined.

* 4.4.8 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Methods of inspection.

4.5.1 High humidity resistance test. Three 4 by 4-inch specimens shall be laid flat, coated side up, on a supporting plate and the assembly placed in a desiccator containing water in the lower portion. The water level shall be approximately 1 inch below the specimens. The lid of the desiccator shall be put in place and the desiccator placed in a circulating air oven having a temperature of 125° ± 2°F for a period of 7 days. At the end of the aging period, each specimen shall be removed from the desiccator, visually examined for colorfastness, and then tested immediately in accordance with Method 5512 of FED-STD-191 with the water pressure being applied to the uncoated side.

4.5.2 Insect repellent resistance test. The insect repellent for use in this test shall conform to type I of O-I-503. Three drops of the repellent shall be placed in the center of a 4 by 8-inch sample of coated fabric with the repellent contacting the coated side. The sample shall be folded to form a 4 by 4-inch square with the coated sides contacting each other. The folded sample shall then be placed between two 6 by 6-inch glass plates and a 4-pound weight placed on the assembly and left at standard conditions for 16 hours. The sample shall then be removed from between the glass plates and rated for blocking as required in Method 5872 or FED-STD-191. The results shall be reported as pass or fail.

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4.5.3 Hydrostatic resistance test. The seam and bartack hydrostatic test shall be performed as specified in Method 5514 of FED-STD-191 with the following exceptions:

(a) The liner shall be tested without cutting and one test shall be made on the side seam, one where the tie cord is bartacked on the liner, and two in the peripheral bottom seam.

(b) When testing the seams, the seam shall be positioned in the 4-1/2 inch diameter test area with the outside of the seam (coated side of the liner) contacting the water.

(c) When testing the bartacked tie cord, the bartack shall be positioned in the 4-1/2 inch diameter test area with the water contacting the coated side of the liner.

(d) The water pressure shall be raised to 25 centimeters and held for 1 minute.

(e) Leakage shall be defined as the appearance of water in any portion of the 4-1/2 inch diameter test area.

The presence of one or more leaks shall be considered a test failure.

* 4.5.4 Seam blocking test. One side seam and one bottom seam of the liner shall be tested for resistance to blocking by Method 5872 of FED-STD-191 except that the test specimens shall be prepared by folding the liner so that sealed seam surface is superimposed with the coated side of the fabric on the outside of the liner and with the seam positioned in the center of the glass plate.

* 4.5.5 Adhesion of strapping test. The test specimen shall be a 6 by 3-inch section of side seam area cut from the liner to be tested, with the strapping centered thereon. One end of the strapping shall be peeled back 3 inches to provide a tab. The end of the specimen from which the tab was peeled shall be clamped uniformly across its width and the clamp shall be fastened in a horizontal position so that the specimen may hang freely in a vertical position. A weight shall be attached to the tab by means of a clamp across its width 1 inch from the end of the tab, total of clamp and weight to be 3 pounds. As soon as the weight acts on the specimen which shall hang freely without twist, an inked marking shall be made at the peel line to serve as a gage mark for determining the amount of peel occurring during the test. Allow the weight to hang for 5 minutes, then immediately mark the resulting peel line in the same way. Measure the distance between the two lines to the nearest 1/32 inch and report as the amount of peel.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each liner shall be smoothed out flat with the tie cord tucked inside. The bottom shall be folded up on the wall of the liner and then folded in half in length. Twenty folded liners of one size only, shall be stacked flat, alternately reversed, to form an even bundle. Each bundle of liners shall be securely cross-tied or tied at each end with cotton tape or twine. The three different sized bundles of liners shall measure approximately the dimensions specified in table IV.

TABLE IV. Bundle dimensions, quantity and arrangement of liners per pack, approximate inside dimensions 1/ of shipping containers, and pallet pattern numbers

Size of liner	Approximate bundle dimensions (inches)	Quantity of liners per pack (each)	Arrangement of liners per pack (length x width x depth)	Approximate inside dimensions 1/ of shipping containers (inches)	Pallet pattern number MIL-STD-147
1	10-1/2 x 8 x 5	240	2 x 2 x 3 (flat)	21-1/2 x 16-1/2 x 15	3
2	18 x 12-1/2 x 5	120	2 x 1 x 3 (flat)	25-1/2 x 18-1/2 x 15	90
3	24-3/4 x 15-1/4 x 5	60	1 x 1 x 3 (flat)	25-1/4 x 15-3/4 x 15	3

1/ Approximate inside dimensions are furnished as a guide only.

5.1.2 Commercial. Liners shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Liners of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container having approximate inside dimensions as specified in table IV. The fiberboard shipping containers shall conform to style RSC-L, grade V2c of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15C of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping

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container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.7. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Liners of one size only, preserved as specified in 5.1, shall be packed in the quantity, arrangement, and within a snug-fitting fiberboard shipping container having approximate inside dimensions as specified in table IV. The fiberboard shipping container shall conform to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic , variety DW, grade 275 of PPP-B-636. Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.7.

5.2.2.1 Weather-resistant containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.7.

5.2.3 Commercial packing. Liners, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), liners of one size only, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern number shall be as specified in table IV and in accordance with MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951.

6. NOTES

6.1 Intended use. The liner is intended to be used as a liner for field pack to hold items other than subsistence items that must be protected from moisture. The liner is also intended as a flotation device for the individual carrying the field pack.

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* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size required (see 1.2).
c. When a first article is required (see 3.2, 4.3 and 6.5).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 Samples. For access to sample of the end item, address the contracting office issuing the invitation for bids.

6.4 Light meter. A meter found successful in measuring exterior light is the Weston Illuminator Meter Model 756 or 703, type 8 Illumination Meter, 0 to 30 foot candles (with Viscor filter) made by Daystrom, Inc., Weston Instrument Division, Newark, New Jersey 07114.

* 6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.6 Subject term (key word) listing.

Liner
Field pack
Moisture resistant

* 6.7 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 99

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Air Force - 82
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DLA - CT

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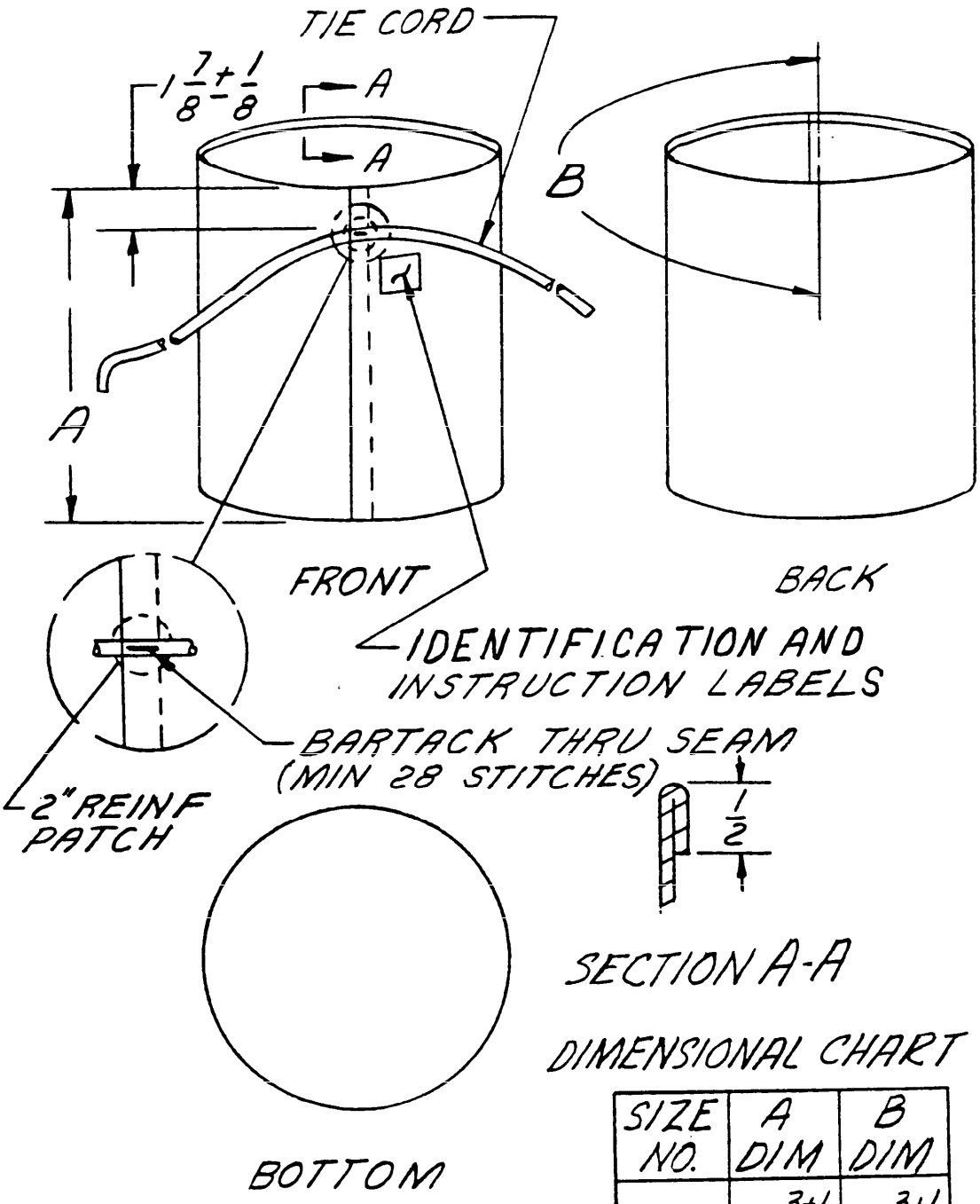
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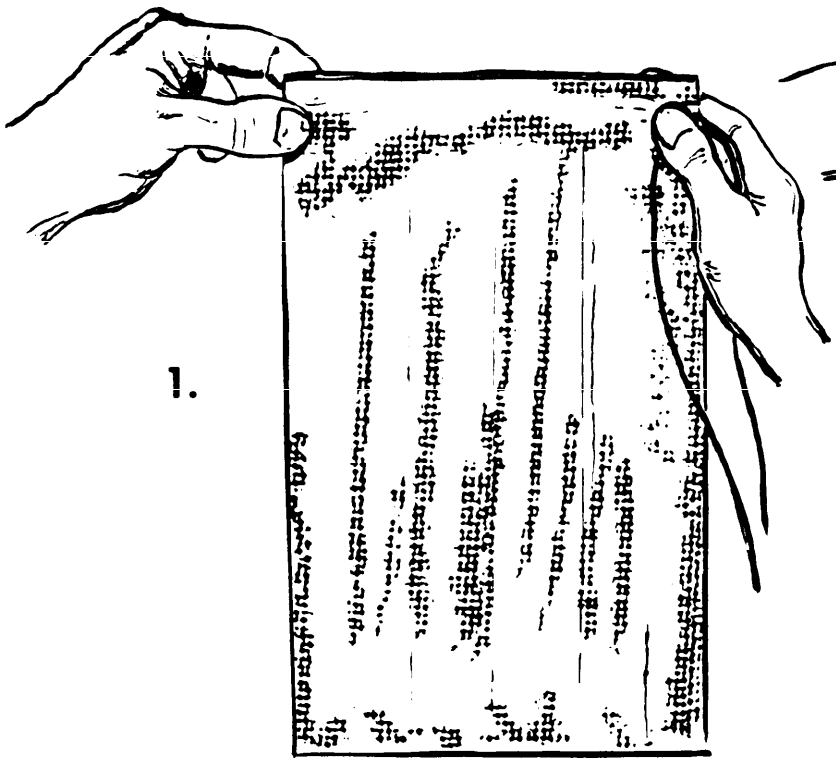
DIMENSIONAL CHART

SIZE NO.	A DIM	B DIM
1	$17\frac{3+1}{4-2}$	$10\frac{3+1}{4-4}$
2	27^{+1}_{-2}	$19\frac{1+1}{2-4}$
3	30^{+1}_{-2}	$25\frac{1+1}{4-2}$

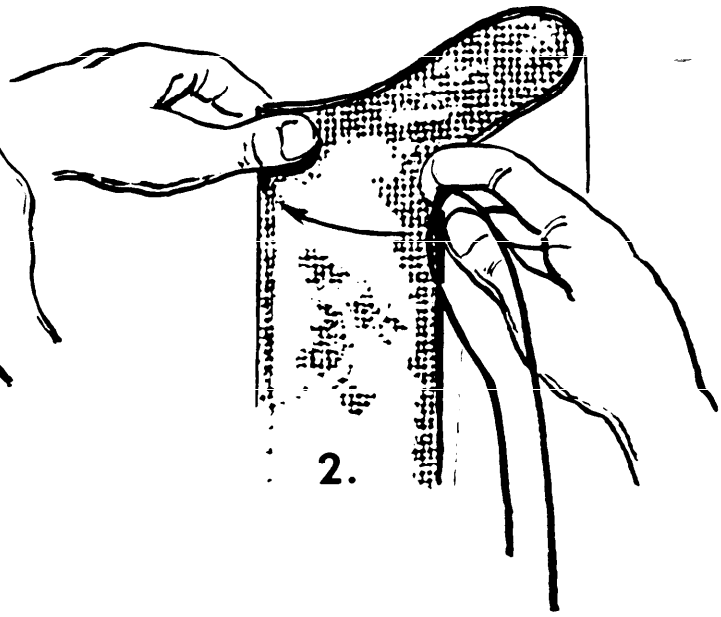
FIG. 1 - LINER, FIELD PACK, SIZES NO. 1, 2 & 3

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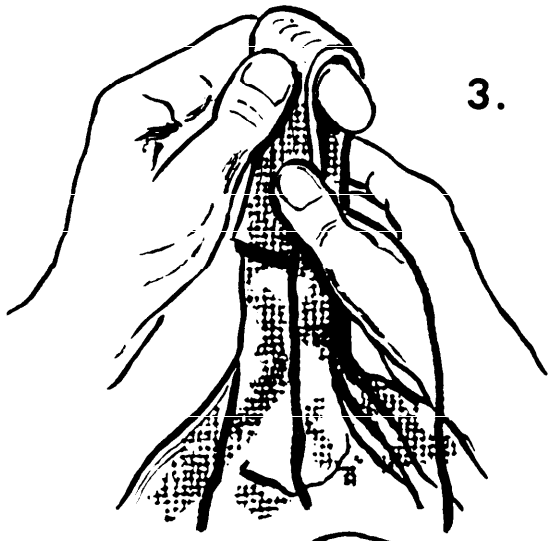
WATERPROOF PACKLINER BAGS



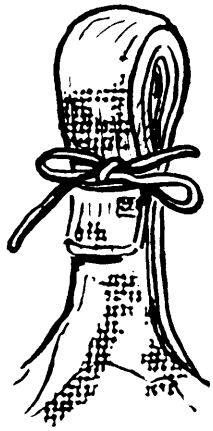
1.



2.



3.



4.

To get tightest closure:

1. Flatten mouth of bag (Fig. 1).
2. Fold flat in half (Fig. 2) and repeat 4 times (approx). Then insert finger to make opening and inflate about 50% of air.
3. Bend over top 3 inches (Fig. 3) on folded material keeping tape on outside.
4. Wrap tape tightly around folded material and tie bow knot.

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FIGURE – 2

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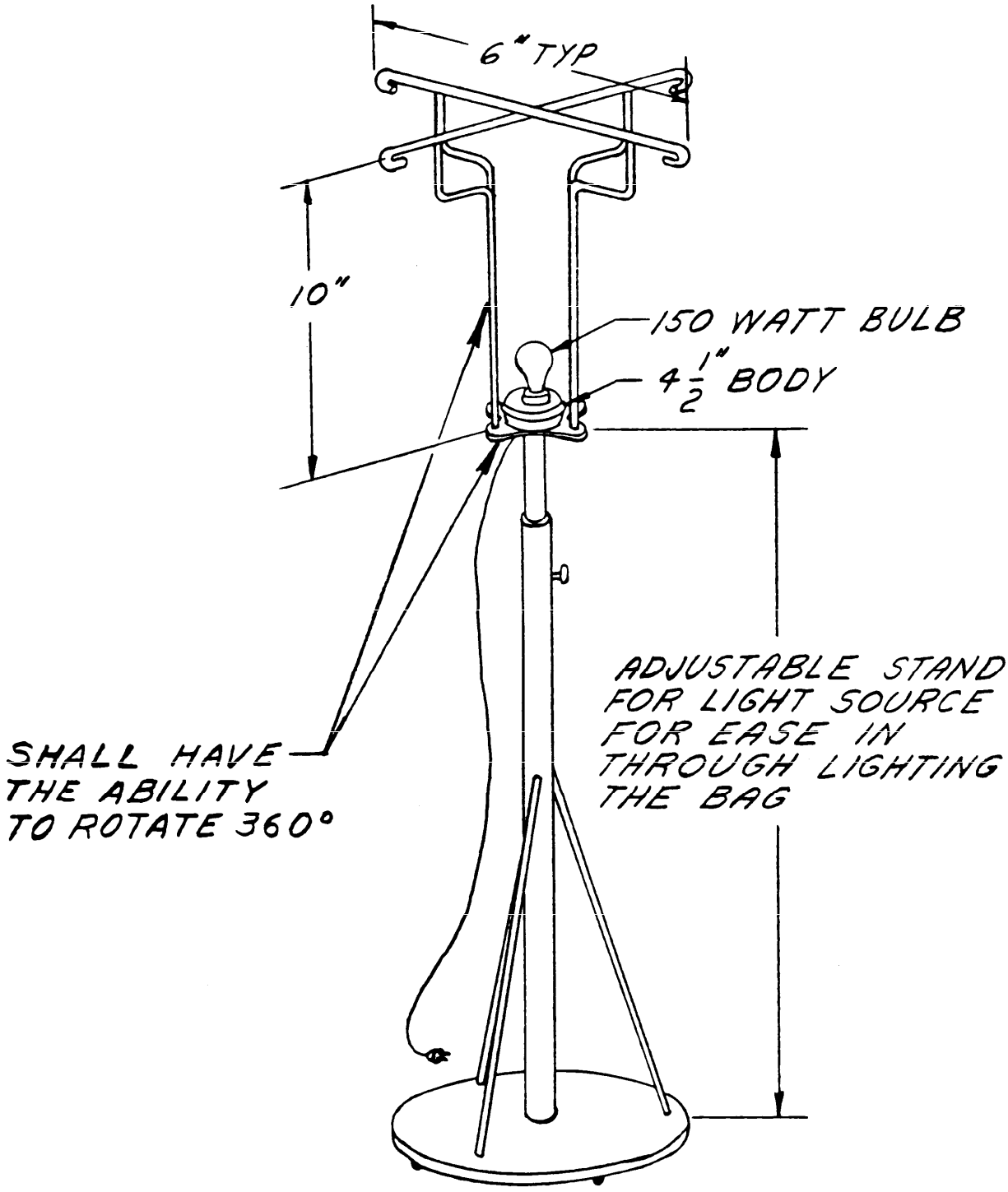


FIG. 3 THROUGH LIGHTING DEVICE FOR SIZE 1, 2 & 3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-43720C	LINER, FIELD PACK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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